Gesture control of digital screen using Kinect V2

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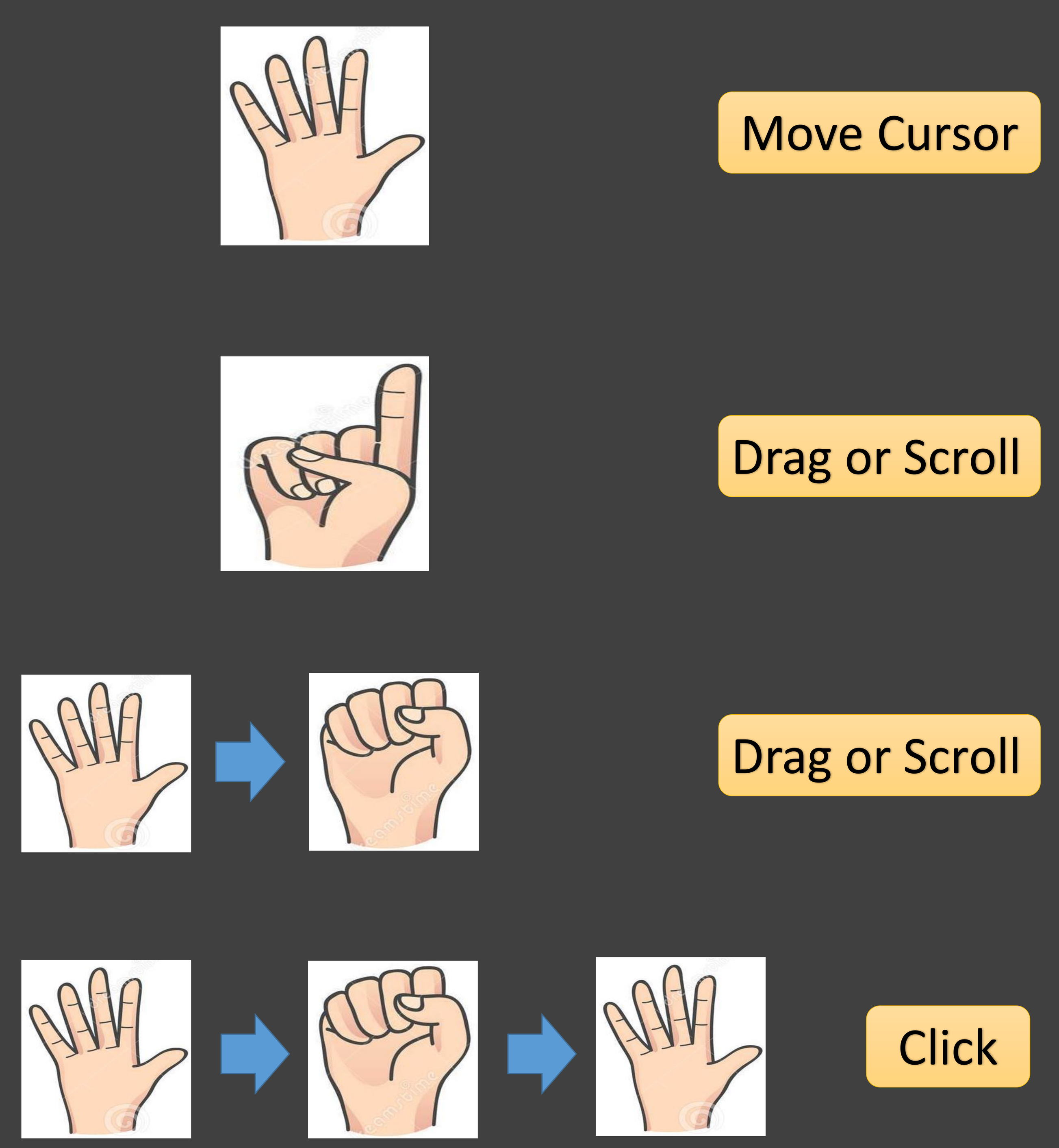
Motivation

The TV display outside the PQRC in Valentine Hall does not draw much attention. It is a non-interactive display that people cannot select what they want to see. This is a common problem for many electronic displays that are installed in airports and malls.

Gestures and Cursor Control



Purpose and Benefits

Purpose

Use Kinect V2 to create a program that allows people to control the unreachable displays such as ads displays in public scenario, and also adds a power-saving feature to them. The method of control needs to be intuitive, simple, and familiar.

Benefits

- Be able to control unreachable electronic displays.
- Turn on/off the screen based on usage to make the display more power-saving.
- Be more friendly to disabled people and cheaper than regular touchscreen.
- Be more sanitary via touchless gesture control.

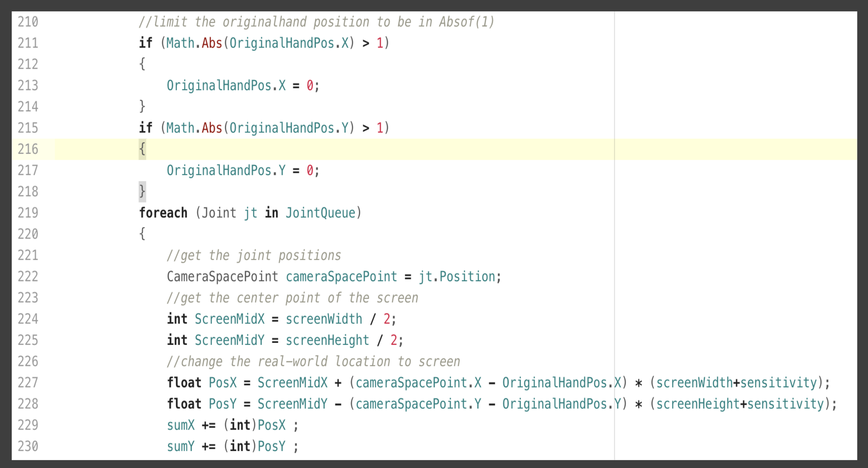
Method: Cursor Control

❖ **Why choose cursor control?**

- Click, drag, and scroll are basic controls for browsing purpose in public scenario, and map nicely to familiar cursor control on desktop and laptop computers.
- It is intuitive and simple for people who have never use gesture control before.
- Visual Gesture Builder does not provide the necessary precision for sensitive control.

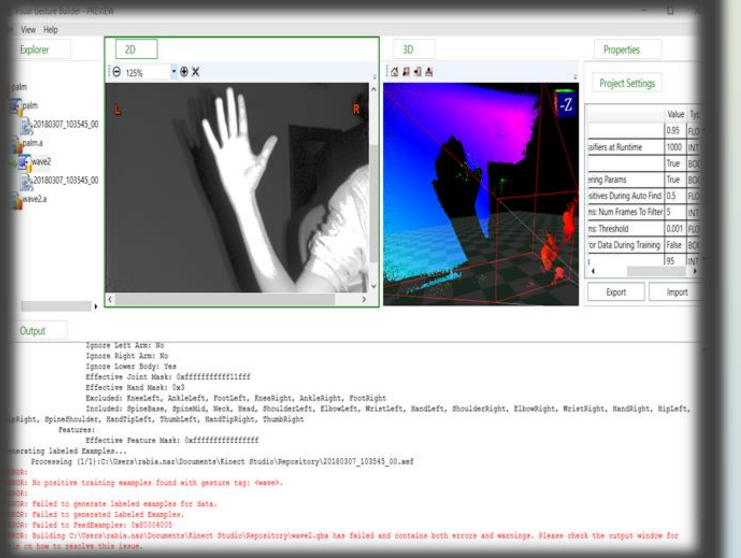
❖ **How is it implemented?**

- There are DLLs(Dynamic-link Library) that set and retrieve the cursor position and enable/disable displays.



- To prevent the cursor move away while doing the hand close-open, we assign weights on wrist and elbow positions when updating the cursor coordinates.
- Stored a series of hand positions into a queue and calculate their average to reduce jitter. We implement linear interpolation to smoothen the cursor movement.

Hardware and Software



- Kinect V2: depth/gesture camera
- Visual Studio: C# Integrated Development Environment
- Kinect SDK 2.0: Software development Kit.
- Visual Gesture Builder: Record gesture videos and uses machine learning models to train these gestures.

Limitations and Future Work

❖ Limitations:

- The skeleton detection by Kinect is not always reliable and can be affected by non-operators around the operator.
- Kinect sometimes cannot capture a hand-close gesture if it is not standard, or if the hand is not facing the camera.
- The cursor still moves when performing the click gesture.
- The lack of resources and out-of-date information due to Microsoft discontinuing development of Kinect.
- Visual Gesture Builder (VGB) is not precise enough for practical use. One needs to record a gesture carefully to catch the operator performing the recorded gesture.

❖ Future Work:

- In this research, I hard-coded for gesture recognition. Machine learning algorihms such as CNN Adaboost and Xgboost can improve its performance by using a large dataset of gesture videos.

Acknowledgement

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